# TRUSTWORTHINESS IN IOT: DISCUSSION FROM WATERLAB AND SMART FACTORY PERSPECTIVES

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**Abstract.** Trustworthiness in Internet of Things (IoT) systems is essential for maintaining critical infrastructure safety, security, reliability, resilience, and privacy. The paper discusses how Savonia University of Applied Sciencese improve IoT trustworthiness through WaterLAB and the Smart Factory. WaterLAB focusses on secure water distribution using Beckhoff PLCs, OPC UA integration, and AI-driven anomaly detection, while the Smart Factory employs Siemens PLCs and OPC UA to ensure reliable industrial processes. The AIQUSEC project implements Prosys OPC UA Forge and SSH PrivX to provide quantum-safe encryption and machine learning solutions. Overall, these initiatives demonstrate scalable and trustworthy IoT systems in accordance with ISO/IEC TS 30149.

**Keywords:** Trustworthiness, Smart Factory, WaterLAB, OPC UA, Quantum-Safe Security, Machine Learning, ISO/IEC TS 30149

## 1 Introduction

Trustworthiness is the level that a system operates as intended with environmental disruptions, loss of performance quality and accuracy, human errors, system malfunctions, and ex-ternal attacks [1]. Trustworthiness is characterised by key principles such as safety, security, privacy, reliability, and resilience which are essential for IoT systems to function within larger industrial ecosystems (see Fig. 1). For the Internet of Things (IoT), trustworthiness is essential to ensuring the security and reliability of systems, particularly in critical infrastructures like manufacturing and water management where any failure could have a disastrous impact on public safety, the environment, and financial stability. It reduces the risks caused by cyberattacks, system failures, and operational interruptions by ensuring that IoT systems are designed to function normally even in the context of these challenges [2].

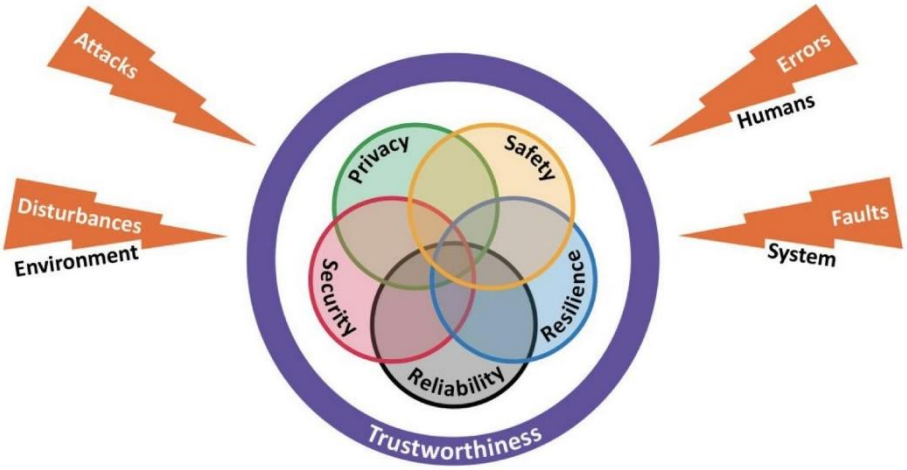


**Fig. 1.** Trustworthiness Characteristics [1]

The AIQUSEC (AI-based quantum-safe cybersecurity automation and orchestration for edge intelligence in future networks) project is one of the research projects at Savonia University of Applied Sciences that focusses on this idea of trustworthiness. AIQUSEC aims to improving IoT security by integrating AI-driven cybersecurity solutions with quantum-safe technologies that are suitable for edge computing environments in industries like smart factories and water distribution systems. The project discusses the challenges of cyber threats, such as the necessity of secure communications, device authentication, and sensor security, all of which are essential for maintaining the integrity of critical infrastructure [3].

Savonia WaterLAB is important for the testing and validation of IoT systems for water distribution which simulate real-world scenarios to assess cybersecurity measures and secure system resilience. The WaterLOOP system within WaterLAB offers a controlled environment for finding and fixing vulnerabilities which is essential for maintaining the trustworthiness of water management systems. Like this, Savonia Smart Factory serves as an experimental environment for cybersecurity analysis and implementation which ensure the security and resilience of industrial IoT systems.

The following article will explore at how Savonia University of Applied Sciences contributes to IoT trustworthiness, specifically the AIQUSEC project, WaterLAB, and Smart Factory, and how they are related with global standards like the ISO/IEC 30149 IoT trustworthiness framework.

# 2 Key Characteristics of IoT Trustworthiness

The notion of trustworthiness in Internet of Things systems has several aspects and guarantees that systems operate safely, securely, and reliably in a variety of conditions. (ISO/IEC TS 30149, 2024) Trustworthiness in IoT systems is defined by the following key characteristics, particularly in industrial environments:

## 2.1 Safety

Safety helps to guarantee the IoT system is not harming people, property, or the environment. It is established by a set of safety goals that describe the circumstances in which a system should function safely. To address safety issues, predictive analyses of data from IoT devices are usually used. This enables the system to detect possible faults and take action to fix them before problems happen. Safety is essential in factory environments to avoid irresponsibility or unexpected system malfunctions that might result in accidents or disastrous incidents. To ensure safety, generic program properties are used. The safety property explains that nothing harmful occurs during system execution, indicating that the system will not enter a bad state. The liveness property claims that something positive will occur, indicating that the system will eventually achieve a good state. These features guarantee that the system avoids failure situations and operates in a reliable and safe way [4].

## 2.2 Security

Security is the shield that keeps an IoT system safe from malicious attacks, illegal access, and data breaches. It includes safeguarding data and making sure that applications, services, and control systems are protected from attack. IoT security includes both intentional (cyberattacks) and unintentional (malware) threats. The objective is to reduce these risks by creating suitable security measures. Risk assessments can be utilized to create a comprehensive threat model that detects potential vulnerabilities and develops actions for both intention-al and unintentional security attacks. Security objectives usually follow the CIA triad (Confidentiality, Integrity, Availability) and include additional characteristics of trustworthiness, including attack resilience, safety, re-source integrity, fail-safety, and privacy. Security objectives should be precise, dependable, and in line with the purpose and desired usage of the system. All stakeholders should be included in the process of design, and security must be considered from the beginning to make sure the system is secure against attacks and protects data and privacy [4].

## 2.3 Reliability

In IoT systems, reliability is the ability of the system to perform intended tasks for a certain period under specified circumstances. It is essential for maintaining continuous operations, particularly in critical infrastructure where failures might have devastating consequences. The goal of reliability is to keep the system operating consistently

throughout time whatever the environmental changes, network outages, or damage from use. Reliability is important in areas where IoT devices control or monitor critical infrastructure, such as manufacturing and water management, where any interruption might have serious operational and financial consequences [4].

### 2.4 Resilience

In the context of Internet of Things systems, resilience is the ability to provide an appropriate level of service during times of unexpected disturbances such system failures, power outages, or cyberattacks. In contrast to reliability, which ensures functionality within normal conditions, resilience focused on the ability of the system to recover from failures and continue functioning despite challenge. Key features of resilience include Redundancy, Adaptability and Recovery. Resilience is important in IoT systems which support critical infrastructure because unexpected incidents can cause major operational failures if the system is not built to recover rapidly [4].

### 2.5 Privacy

Privacy refers to the protection of individuals personally identifiable information (PII) from unauthorized access or usage. Privacy in IoT systems refers to ensuring that personal information is only collected, processed, and shared with authorized parties and that privacy regulations like the General Data Protection Regulation (GDPR) are followed according to. Privacy goals emphasize Unlikability, Transparency and Intervenability. To guarantee that data protection is continuously maintained and that users have control over their personal information, privacy should be included in the design and lifetime of IoT systems. Regular privacy risk assessments and the use of privacy management technologies are also necessary to maintain data protection standards throughout the IoT ecosystem [4].

## 3 Trustworthiness Framework in Savonia

### 3.1 Savonia WaterLAB

Savonia WaterLAB is a specialised research and development facility that focusses on system control and water management. Its infrastructure is built on advanced technology that follows to the principles of trustworthiness, such as resilience, security, privacy, and reliability.

**Beckhoff PLC.** The Beckhoff PLC (Programmable Logic Controller) technology serves as a basis for WaterLAB's automation and monitoring system. It ensures that the water management system of the lab functions properly by monitoring multiple operations. WaterLAB ensures system reliability and security by integrating Beckhoff's advanced industrial controllers. The real-time functionality of the system is essential since

any failure could influence the quality of the research and create operational interruptions. The PLC system aims to meet the standards for trustworthiness included in ISO/IEC 30149 such as reliability and resilience by employing real-time monitoring, fault detection and predictive maintenance.

**OPC-UA Integration.** The OPC UA (Open Platform Communication for Unified Architecture) integration in Savonia WaterLAB is important for ensuring the secure and reliable transmission of data between systems and devices. OPC UA is part of the IoT trustworthiness architecture which allows for safe data transfer and system interoperability in real-time water system monitoring. In industrial systems, OPC UA provides an efficient, cross-platform communication protocol that ensures data security and reliability. The OPC UA Security Model uses advanced security techniques including integrity checks, encryption, and authentication [5]. It avoids possible security risks including denial of service attacks, message tampering, and unauthorised access. OPC UA security features support CIA triad (confidentiality, integrity, and availability) and data non-repudiation as described in the OPC UA Security Model (ISO/IEC 62443). OPC UA Safety is used to guarantee the transfer of safety-critical data, particularly in situations where there are communication errors or system breakdowns [6]. According to ISO/IEC TS 30149:2024 safety criteria, this ensures that the system maintains its safe state even in the event of disturbances.

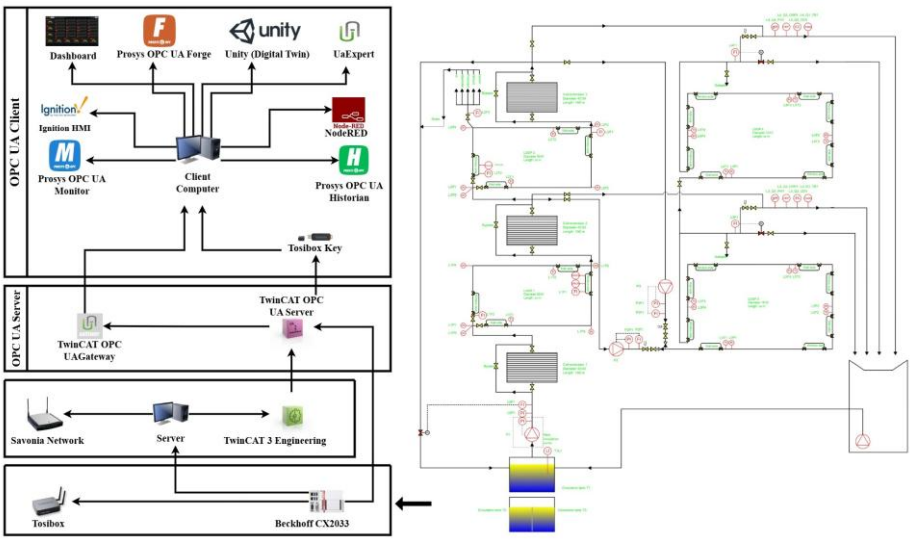


**Fig. 2.** WaterLOOP OPC UA

In WaterLAB (see Fig. 2), OPC UA guarantees seamless communication between a variety of devices including PLCs, sensors and control systems. This facility increases operational resilience by implementing OPC UA cross-platform architecture that

allows for real-time monitoring and quick responses to system anomalies which enhance the reliability of water management systems.

**AI-driven Cyber Security and Anomaly Detection.** In Savonia WaterLAB, anomaly detection and AI-driven cybersecurity are necessary components for maintaining the reliability and trustworthiness of IoT-based system. WaterLAB implements machine learning algorithms that continuously monitor for abnormal behaviours or possible security breaches as part of its adherence to securing its infrastructure. This system has been designed to predict and react to anomalies if they are caused by internal system malfunctions or external cyberthreats which guarantees the integrity of the water management system. To validate this framework, we analysed the performance of the LSTM-VAE model on the WaterLAB datasets with 20, 62, and 69 sensors, respectively. Our analysis found that the LSTM-VAE model works optimally and effectively in fulfilling the anomaly detection requirements, with an F1 score of around 0.9.

According to ISO/IEC TS 30149, The integration of AI-driven cybersecurity in WaterLAB aligns with the characteristics of IoT trustworthiness. The anomaly detection features of the system also meet the security characteristic describe in the specification which guarantees the confidentiality, availability and integrity of that system. Additionally, the AI-driven method improves resilience by avoiding major operational failures and responding to unexpected events. WaterLAB improves its capacity to operate safely and securely by highlighting reducing risks, quick reaction, and continuous monitoring. This shows a proactive approach to risk management and satisfies stakeholders expectations for trustworthiness.

### 3.2 Savonia Smart Factory

Smart Factory in Savonia University of Applied Sciences (Lucas-Nülle Smart Factory) is a good example of Industry 4.0 principles, integrating advanced IoT, automation, and security technologies to optimise manufacturing processes. With its highly automated, secure, and efficient production processes, the Smart Factory environment is an important environment for creating the next generation of industrial systems.

**Siemens PLC.** The basis of the Smart Factory is the Siemens PLC system, which serves as the production process control layer. The PLC system manages production phases and integrates with MES (Manufacturing Execution System) and ERP (Enterprise Resource Planning) systems to improve factory operations adaptability and scalability. Specifically, Siemens S7-1200 PLCs are used for local management of production stations, while a Siemens S7-1500 Master PLC coordinates the system and maintains factory communications. These PLCs manage the operation of machines, conveyor belts, and quality control procedures and provide accurate, real-time automation. Profinet is a high-performance industrial Ethernet protocol that facilitates communication between

PLCs and devices and ensures safe and quick data transfer throughout the factory line. Communication between the MES/ERP and PLC systems is established using CoAP (Constrained Application Protocol) to ensure the real time pro-duction data. (see Fig. 3)

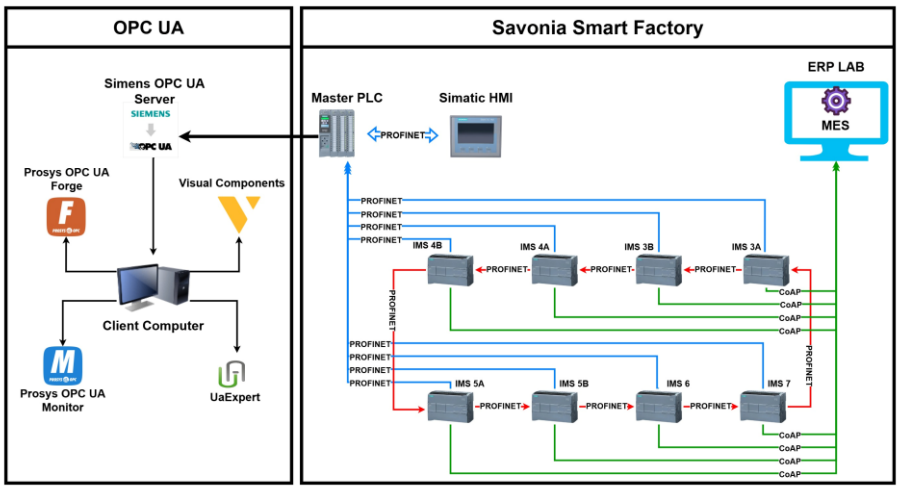


**Fig. 3.** Savonia Smart Factory Communication Architecture

The integration of Siemens PLCs with industrial IoT systems enables continuous system performance monitoring, predictive maintenance, and real-time decision making. This improves the reliability and resilience of the factory and in line with the ISO/IEC TS 30149 criteria for industrial IoT systems.

**OPC UA Integration.** Like WaterLAB, the integration of OPC UA within the Savonia Smart Factory aims to improve the system interoperability and ensure effective communication between devices from different manufacturers. The OPC UA protocol makes data transmission easier between PLC systems and higher-level software platforms like Visual Components, which contains a dynamic digital model of the factory. Because of this connectivity, digital models and physical production systems get synchronised in real time which allow operators to monitor and make necessary adjustments. OPC UA security features in this setup also aligns with the ISO/IEC TS 30149, which highlights the importance of resilient and secure communication frameworks in industrial IoT systems. By using OPC UA, the Savonia Smart Factory makes sure that its production procedures are safe, scalable, and future-proof and enable it to incorporate new technologies and adjust to changing market demands.

**AIQUSEC Lab.** The AIQUSEC Lab is a collaboration between Savonia University of Applied Sciences and Jyväskylä University (JYU) that focusses on and creating quantum-safe cybersecurity solutions and using machine learning for anomaly detection (Fig. 5). A key component of this security lab is the implementation of Prosys OPC UA Forge combined PrivX, a privileged-access management solution from SSH which together aims to improve security across Savonia Smart Factory.

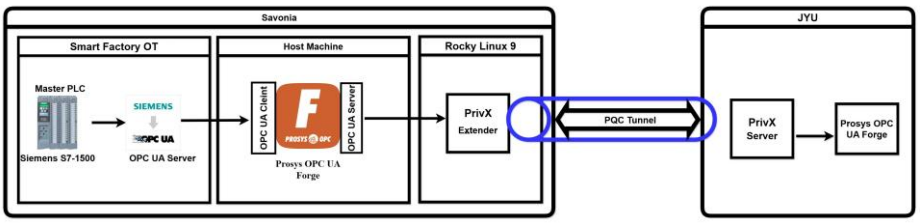


**Fig 5.** AIQUSEC Lab Architecture

Prosys OPC UA Forge is an important part of the setup and served as a secure and scalable gateway for connecting Savonia Smart Factory setup to external IT systems. It serves as both an OPC UA client to the Siemens S7-1500 Master PLC for collecting operational data from the manufacturing stations in real-time, and an OPC UA server to safely transmit this unified data to external systems. Its several features such as unified address space, data harmonisation, support for multiple industrial protocols (e.g., Modbus, S7comm, EtherNet/IP) and connectivity via MQTT and REST provide an entire solution for secure industrial data transfer [7].

To enable quantum-resilient remote access, the lab implements SSH's PrivX Extender and PrivX Server. The PrivX Extender is deployed on a Rocky Linux 9 virtual machine in the Smart Factory to establish a secure outbound tunnel to the PrivX Server at JYU. This tunnel employs post-quantum cryptographic (PQC) methods with TLS-based end-to-end encryption to ensure long-term security against quantum-enabled threats. PrivX removes the need for static credentials by using short-lived certificates and immediate authentication which complies with zero-trust principles [8].

On the JYU side, researchers and security specialists can securely access Smart Factory data using Prosys OPC UA Forge without revealing critical infrastructure to the internet. Later JYU will use machine learning models to analyse real-time data from the Smart Factory to detect anomalies. By facilitating the early detection and mitigation of security risks, this integration improves overall security, resilience, and reliability of the factory.

Aligned with the ISO/IEC TS 30149 IoT trustworthiness framework, the AIQUSEC Lab combines quantum-safe encryption with machine learning-driven anomaly detection which enables early detection of cybersecurity vulnerabilities. This combined approach increases the resilience and dependability of smart manufacturing processes,

transforming the lab into a dynamic platform for secure digital transformation in Industry 4.0.

## 4 Collaborations with University and Industries

The AIQUSEC project includes a few key industrial partnerships with contributions from partners in the domains of operational technology (OT), smart manufacturing, and cybersecurity. These collaborations are important for achieving the projects objectives and guaranteeing that the research findings are relevant and helpful in real industry environments. Jyväskylä University (JYU) plays an important role to the AIQUSEC project, particularly in the development and implementation of machine learning models for anomaly detection and developing AIQUSEC LAB. SSH offers in quantum-safe solutions with the PrivX product which is utilised in the Savonia Smart Factory for secure privileged access management. Prosys contributes to the project by offering different OPC UA products (Prosys OPC UA Monitor, Prosys OPC UA Forge) that are tested and implemented at both the Savonia Smart Factory and the WaterLAB. Each of these partnerships advances the goal of the AIQUSEC project to develop scalable, efficient cybersecurity systems that incorporate secure communication protocols like OPC UA, AI-driven anomaly detection, and quantum-safe technology.

## 5 Future Recommendations

The AIQUSEC project should continue improving quantum-safe encryption, advancing AI-based anomaly detection with JYU, and advancing the integration of OPC UA protocols along with Prosys to provide secure and interoperable data transfer. Strong collaborations with JYU, Prosys, SSH and other industry specialists will be important in achieving success in cybersecurity, IoT, and digital evolution. At the same time, Savonia must test and validate its cybersecurity solutions on a regular basis using simulated attacks and vulnerability assessments to detect vulnerabilities early, maintain ISO/IEC TS 30149 compliance, and ensure operational resilience. Both data protection and scalability should remain priority. Strong security measures, such as encryption, access control, and secure storage of sensitive IoT data, must be included into the system architecture from the beginning. As industrial demands grow, both the Smart Factory and WaterLAB should use modular, scalable architectures that allow for efficient scalability while also maintaining reliability, security, and future development.

## 6 Conclusion

The successful implementation of IoT technology in WaterLAB and the Savonia Smart Factory shows an effort to enhancing operational efficiency and securing critical infrastructure. These solutions fulfil the increasing demand for reliable and secure smart IoT systems in industrial environments by using advanced technologies to provide seamless data transfer, robust security, and system resilience. These initiatives establish Savonia

as an example of using IoT to deal with difficult challenges in water management and industry.

As Savonia moves forward, its dedication to ongoing collaborations in the industry and following global standards like ISO/IEC TS 30149 will be important for dealing with current challenges of cybersecurity, scalability, and data protection. Savonia will continue improving its infrastructure with regular innovation and research, ensuring that the WaterLAB and Smart Factory remain safe and adaptable to future demands. The integration of these technologies not only increases operational capabilities but also shows how important trustworthiness is to the sustainability and security of IoT-based infrastructures.

## 7 Acknowledgments

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